CHE 321 – Chemical Reaction Engineering
Fall 2019

Homework Assignment – 5 (20 points)

Due: 12 PM, November 18th 2019

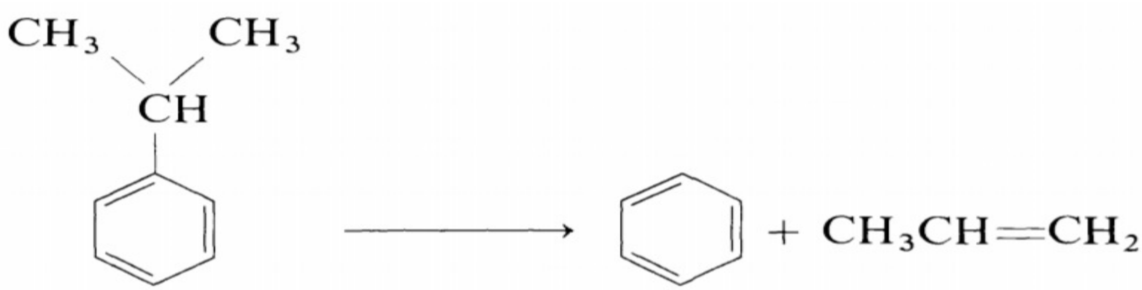
Instructions

1) Email the answers, figures and explanations as a single pdf file to TA. Also submit the MATLAB codes.
2) Make necessary assumptions.
3) 5 points will be deducted for submissions after the deadline.

Objectives

1) To study the effect of simultaneous mass transfer and reaction in a catalyst pore.
2) To understand the concept of effectiveness factor.

Consider the cracking reaction of Cumene into Benzene and Propylene which is given by



The reaction occurs in the presence of silica-alumina catalyst isothermally at 750 K and the specific volume remains constant. Assume the reaction follows first order kinetics which is given by

$r' = k' C_s$ where r' is the rate of the reaction per unit surface area (mol/m^2.s), k' denotes the rate constant(m/s) and C_s denotes the concentration of cumene at the catalyst surface (mol/m^3). All the pores are assumed to be uniform and they remain cylindrical of constant radius r_o and length L. The reaction occurring on the catalyst is limited by the diffusional mass transport into the pore. D denotes the diffusivity of cumene.

1) Write mole balance inside the cylindrical pore assuming steady state and provide appropriate boundary conditions. (**5 points**)
2) Solve the mole balance equation with respective to boundary conditions analytically and express the concentration of cumene at the surface as a function of length of the pore and Thiele modulus defined as $\phi = L(2k'/r_oD)^{1/2}$. **(8 points)**
3) Find the expression for the effectiveness factor, η in terms of Thiele modulus ϕ. (**5 points**)

CHE 321 – Chemical Reaction Engineering
Fall 2019

4) Plot the effectiveness factor as a function of the Thiele modulus (0 to 5). (**2 points**)

Bonus points – 20

Repeat the same problem for LHHW (Langmuir Hinshelwood Hougan Watson) kinetics given by, $r' = k' C_s / (1 + K C_s)$.